

Comparative Analysis of Interpretability and Feature Combinations across Dementia Classification Models

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Abstract—Early and accurate detection of dementia through linguistic analysis has become a compelling application of artificial intelligence in healthcare. With the advent of Deep Learning and Natural Language Processing, identifying whether an individual is cognitively impaired or cognitively sound through text classification has recently seen an emergence in the domain of both medical science and computer science. This dissertation investigates how machine learning models interpret and prioritise feature combinations for dementia classification, using the DementiaBank dataset. Four distinct model types Support Vector Machines (SVM), Random Forests, and Bilinear Long Short-Term Memory (BiLSTM) networks are analysed through the lens of feature attribution methods. Both model-agnostic tools (e.g., SHAP, LIME) and model-specific techniques are employed to compare how each model highlights important combinations. The study evaluates interpretability of feature combinations across these models through a heuristic approach, with particular attention to lexical, syntactic, and discourse-level features. The findings aim to support the development of explainable systems for real-world healthcare decision-making.

Index Terms—Dementia, Feature Combinations, Text Classification, Interpretability

I. INTRODUCTION

DEMENTIA is a neurodegenerative disorder associated with the decline of cognitive functioning. The term is not specific to a single disease and covers conditions such as Alzheimer’s disease, vascular dementia, Lewy body dementia, and Frontotemporal dementia. Among these, the most common one is the Alzheimer’s disease in the United Kingdom (Alzheimer’s Society, 2025). In 2012, the number of people living with dementia was estimated to be 800,000, and it is predicted to grow up to 1 million by 2025 (Alzheimer’s Research UK, 2025). In his report, Farrar, 2024 predicts the cost of dementia to rise to £200 billion by 2025 from £12 billion in 2012. The cure for dementia remains undiscovered but a timely diagnosis of the disorder may provide opportunities for both the patients and the medical experts to take the necessary preventive measures. Recent development in AI in the fields of Natural Language Processing and Computer Vision have been finding their way in the healthcare sector to diagnose the symptoms of neurodegenerative disorders. Some key indicators associated with cognitive decline are linguistic

abilities such as reduced lexical diversity, increased hesitations, and syntactic errors (Kaltsa et al., 2024).

The aim of this study is to determine (1) whether feature combinations are consistent across the dementia models, and (2) how interpretability differs between classical and deep learning approaches. The feature combinations we will be investigating, are combinations of text features, which is the primary information based on the transcripts, and numerical features, which were curated based on the annotations from these transcripts. We will be comparing the performances and importance of different feature combinations of Support Vector Machines (SVM), Random Forests, and Long Short-Term Memory (LSTM) networks used for diagnosing dementia using DementiaBank dataset. The study focuses on evaluating how each model identifies and prioritises influential input feature combinations, with the objective of understanding discoveries in interpretability mechanisms and feature relevance across traditional and deep learning approaches in dementia classification tasks.

II. RELATED WORK

Recent literature has explored using speech and language data for automated dementia detection. Earlier studies employed handcrafted features with traditional ML classifiers. Orimaye et al. (2014) explored the use of syntactic and lexical features from verbal utterances to diagnose Alzheimer’s disease and related dementias. Using the DementiaBank corpus, they applied machine learning models to distinguish affected individuals from healthy controls, achieving an F-measure of 74% on Support Vector Machine (SVM). Their work highlights the diagnostic potential of lexical and syntactical features and provides a strong precedent for language-driven approaches to neurodegenerative disease detection.

Likewise, Jahan et al., 2024 assessed performances of SVM, KNN, Random Forest, and Artificial Neural Networks using TalkBank’s DementiaBank dataset to examine early signs of dementia. Their approach combined linguistic features such as hesitation markers, word frequency, and syntactic ratios with word and sentence embeddings. Among the models tested, Random Forest and ANN achieved the highest accuracy, exceeding 99%. The study underscores the diagnostic potential of speech-based

machine learning and highlights the value of nuanced linguistic feature engineering.

More recent work has applied deep learning models. Saltz et al., 2021 conducted a study to detect dementia by analysing linguistic features in casual and semi-structured speech. They applied five Transformer-based models including BERT, ALBERT, XLNet, RoBERTa, and ELECTRA on datasets from University of Washington interviews, DementiaBank’s Cookie-Theft corpus, and the ADReSS challenge. Their models achieved F1-scores up to 90% on augmented ADReSS data and demonstrated strong performance across datasets. The findings suggest that Transformer-based NLP models can effectively identify cognitive impairment from naturalistic speech, supporting flexible, real-world diagnostic settings.

Bouazizi et al. (2024) proposed a novel LSTM-based approach for dementia detection using transcribed speech from the DementiaBank Pitt corpus. Instead of relying on language models, they tracked subjects’ shifting focus across predefined image regions during a picture description task. Their model achieved 83.6% accuracy at the subject level, outperforming traditional perplexity-based methods. The study highlights topic coherence and attention patterns as promising indicators of cognitive decline beyond lexical or grammatical deficits.

While linguistic feature extraction is now a standard in dementia detection, prioritising relevant features and discarding irrelevant features can help in improving model efficiency and robustness. The work done by Rajab et al. (2023) involved conducting a study to identify critical features using Feature Ranking on a dataset consisting of neuropathological features from a Cognitive Function and Ageing Study (CFAS) cohort. By applying seven ranking methods, they consistently identified Braak NFT stage, beta-amyloid burden, and cerebral amyloid angiopathy as top predictors of dementia. Their best-performing model, using eight features, achieved 79% sensitivity and 75% precision, highlighting both the promise and complexity of post-mortem diagnostic classification.

In another study, Bansal et al. (2018) presented a comparative analysis of machine learning algorithms J48, Naïve Bayes, Random Forest, and Multilayer Perceptron for detecting dementia using the OASIS MRI dataset. After applying feature selection via CFSSubsetEval, J48 consistently outperformed other models, achieving up to 99.52% classification accuracy. The authors emphasize the importance of early detection and attribute reduction for improving diagnostic performance. Their findings support decision tree-based models as effective tools for dementia classification, especially when combined with targeted feature selection.

In their work, Kurdi (2024) demonstrated that combining lexical features can significantly enhance Alzheimer’s detection accuracy. By applying multiple feature selection techniques (e.g., ANOVA, Chi-square, ReliefF), the study identified optimal subsets that outperformed prior bench-

marks. A 27-feature combination yielded an F1-score of 0.917 when paired with input-length optimization. These results highlight the diagnostic potential of linguistically grounded features and support their use in scalable, task-general models for cognitive impairment classification.

While predictive performance has improved, there is growing interest in model interpretability. Model-agnostic methods like SHAP and LIME have been used to explain individual predictions in medical NLP tasks. For dementia diagnosis, explainability remains challenging due to the heterogeneity of language symptoms. Ilias and Askounis (2022) investigated explainable Alzheimer’s disease detection using transformer-based models on the ADReSS Challenge dataset. They proposed both single-task and multi-task learning frameworks, incorporating MMSE scores as a multiclass classification task. BERT achieved the highest accuracy (87.5%) in single-task learning, while their multi-task model reached 86.25%. They introduced a co-attention mechanism for interpretability and applied LIME to explain predictions. Their linguistic analysis revealed distinct patterns in pronoun usage, verb tense, and filler words between AD and non-AD subjects, supporting model transparency and clinical relevance.

A recent systematic review noted that few works directly compare interpretability across model families Martin et al., 2023. Our study addresses this gap by explicitly contrasting feature importances from both classical (SVM, Random Forest) and deep models (LSTMs) by combining linguistic and handcrafted numerical features extracted from the transcripts. To our knowledge, this is the first work to quantitatively analyse and compare the feature attribution outputs of multiple dementia classifiers on the same data, using both model-specific weights and SHAP values.

III. METHODOLOGY

This study performs a comparative analysis of feature combinations across multiple dementia classification models and their interpretability.

Data: This work makes use of DementiaBank subset obtained from TalkBank. Established in 2002, TalkBank is a comprehensive collection of multilingual data encompassing diverse domains such as first/second language acquisition, conversation analysis, dementia and so on. DementiaBank subsets consists of audio and video recordings with transcriptions of individuals in various languages. We use the Pitt Corpus subset of DementiaBank, specifically the Cookie Theft picture description transcripts. This corpus contains speech transcripts from 549 participants, from healthy controls, individuals with mild cognitive impairment(MCI) and individuals with dementia. Each transcript is a narrative description of the Cookie Theft picture task.

Models: Four different classifiers namely LinearSVC, Random Forest, and BiLSTMs (using pre-trained

GloVe embeddings), with Attention and without Attention, were trained and evaluated on this corpus. Enhancements were then made to push the accuracies of the model towards 80%. We then applied both model-specific and model-agnostic explainability methods to examine which combinations each model uses for diagnosis. Model-specific analysis includes inspecting the learned weights or importance measures (e.g., SVM model coefficients, Random Forest importances). Model-agnostic analysis uses SHAP (SHapley Additive exPlanations) values calculated via the KernelExplainer to attribute importance to input features (Mosca et al., 2022). We compute weighted scoring based on feature importance values of text features and numerical features to quantify their importance across the models. The insights gained can guide the development of transparent AI tools for early dementia screening. Ilias and Askounis, 2022; Saltz et al., 2021.

The evaluation metrics to be used in the study will include standard metrics for classification models. The metrics for classification include accuracy, precision, and F1 score. The contributions of feature importance and their combinations were analysed through heuristic based approach.

Methodology

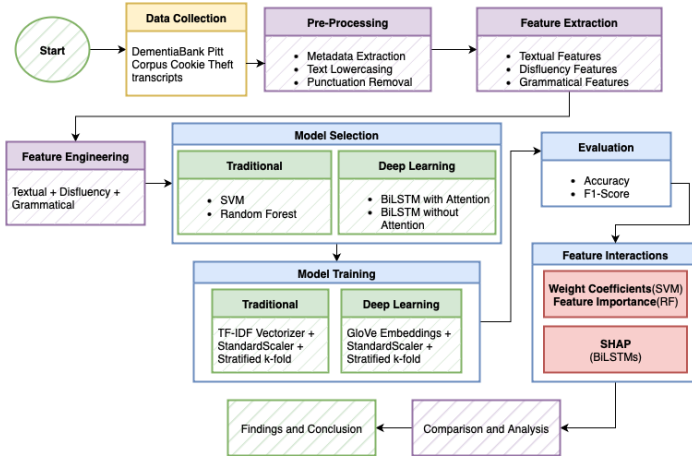


Fig. 1: Project pipeline

1) *Data Collection*: The data used from Pitt Corpus consists of 549 samples from both Control group and Dementia group.

2) *Data Preprocessing*: The Cookie Theft data used for this dissertation are present in CHAT format transcription files(.cha). The data belonging to 'Control' category exist in 'Control' folder and the data belonging to the impaired category, which include 'AD' and 'MCI', exist in 'Dementia' folder.

These files were loaded, read, and metadata, which included age, sex, and diagnosis, and speech data were extracted from them. Once speech or text features were ex-

tracted, text cleaning and normalisation were performed. We then converted the text features to lowercase. Annotated fillers like '&-uh', '&-um' were converted to standard forms like 'uh' and 'um'. This step was followed by removal of punctuations and tokenisation using Natural Language Toolkit(NLTK).

3) *Feature Engineering*: With the pre-processed text, we handcrafted specific features to add nuances to the data based on annotations like [//], [/], +/- and so on, using regular expressions. These included disfluency features like number of repetitions, number of revisions, number of false starts, number of filler words, and number of pauses. Grammatical features which included fragments, subject-verb disagreements and POS bigrams were also created and incorporated.

The handcrafted features are displayed in Table 1. A dataframe is created using text features and numerical features (both disfluency features and grammatical features). The created dataframe consists of 549 samples from 'Control', 'ProbableAD', and 'MCI' categories. One of the drawbacks of the created dataframe, was that many transcripts belonged to the same speakers and were transcribed across different sessions. To avoid this leakage of data, we performed data aggregation based on participant id '@ID'. The finalised dataset consisted of 291 transcripts, with each transcript now belonging to a particular individual.

As this study required classification based on linguistics, we removed attributes 'age' and 'sex' from the dataset.

TABLE I: Handcrafted speech features for dementia detection

No.	Feature	Description	Type
1	Repetitions	Count of immediate word repetitions (e.g., "the the book") indicating disfluency patterns	Numeric
2	Revisions	Number of speech corrections marked by [/] annotations representing self-corrections during production	Numeric
3	False starts	Count of incomplete utterances marked by [/] indicating restarts and hesitation patterns	Numeric
4	Fillers	Total occurrences of hesitation markers including "uh", "um", "erm", "hmm" and variants	Numeric
5	Pauses	Number of temporal disruptions marked by (..) for long and (.) for brief pauses in speech flow	Numeric
6	Unusual bigrams	Count of atypical POS sequences (e.g., DT-DT, NN-NN) reflecting syntactic irregularities	Numeric
7	Disagreements	Number of subject-verb agreement errors detected through POS analysis indicating grammatical inconsistencies	Numeric
8	Fragments	Count of noun-noun sequences identified via POS tagging indicating potential grammatical fragments. A subset of Unusual bigrams	Numeric

The initial dataset also presented three categories of

diagnosis. These included 'Control', 'ProbableAD' or 'AD', and 'MCI'. We plotted the distribution of these categories across the finalised dataset to examine the balance. The plots heavily suggested that there was imbalance between the three classes and a drastic difference between 'AD' and 'MCI', with 167 counts of 'AD' and only 19 for 'MCI'. Since both of these categories are somewhat related based on cognitive decline, statistical tests were performed to determine whether the two categories should be merged or treated separately.

The statistical test we performed is the Mann-Whitney U-test which is non-parametric, suitable for small datasets, and is used for comparing independent groups. We used only numerical features for the groups 'AD' and 'MCI' and performed this test. It was found that out of the eight features, only one feature was slightly significantly different and the remaining seven were slightly and significantly similar. Based on these results, we merged the samples from category 'MCI' and 'AD' to one 'AD' category that included all the samples belonging to impaired individuals. The new dataset had 186 samples/individuals belonging to 'AD' category and 105 samples belonging to control category highlighting imbalance between two different classes.

TABLE II: Mann-Whitney U-test results comparing MCI and AD groups.

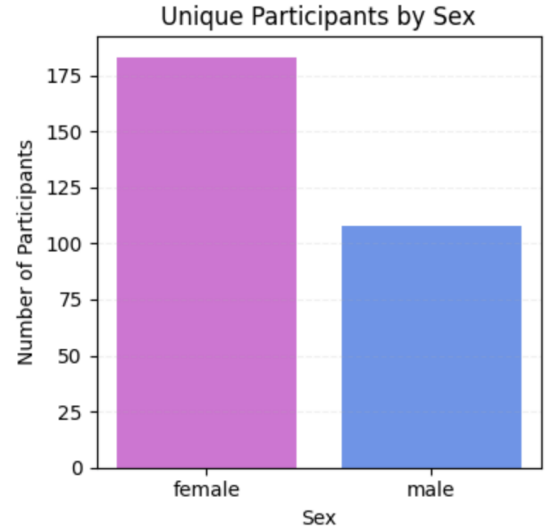
Feature	p-value	MCI mean	AD mean
num_fragments	0.68	6.33	7.63
num_pauses	0.21	0.81	0.63
num_false_starts	0.39	2.35	2.37
num_unusual_pos_bigrams	0.65	5.78	7.42
num_revisions	0.48	1.49	2.21
num_repetitions	0.34	0.43	0.26
num_fillers	0.03	4.12	8.47
num_disagreements	0.41	0.3	0.21

We also compared gender distribution in the dataset as well as gender distribution in each of the two categories using bar graphs, and found that the dataset consisted of 183 females and 108 males, of which 124 belonged to 'AD' category against only 62 males belonging to the same category, and 59 females belonged to 'Control' category against 46 males, suggesting that imbalance may exist due to high number of one gender belonging to one category.

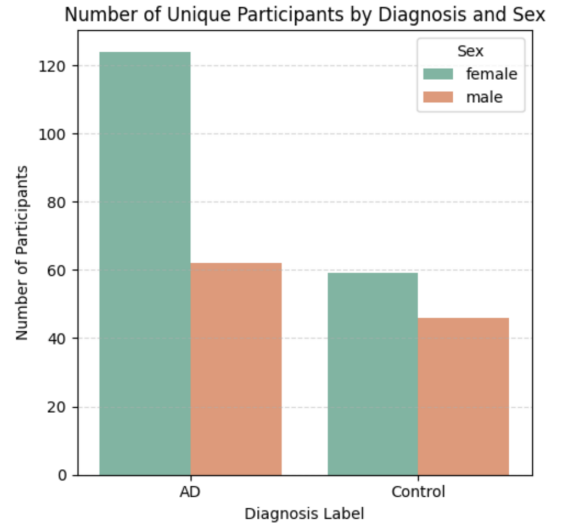
4) *Classification Models*: Four models are implemented to perform classification, namely LinearSVC, Random Forest, BiLSTMs, one with Attention and one without Attention. The models are trained using the training set and are evaluated on test set. Stratified cross-validation is implemented across the four models.

For training and evaluating, we split the dataset into 80% training and 20% test sets. For LinearSVC and Random Forest, the pre-processed text data were transformed into numerical feature vectors using TF-IDF vectorization. Tf-Idf was chosen as text vectorisation based on the results of experiments performed by Cahyani and Patasik, 2021.

In their papers, Wang, 2023 and R et al., 2025 show empirical evidence of using GloVe embeddings with BiL-



(a) Overall distribution by sex



(b) Distribution by diagnosis and sex

Fig. 2: Participant demographics showing gender distribution

STMs for text classification, which we decided to choose for the deep learning setup for this project.

Across all the four models, we implemented Stratified K-fold cross-validation technique to prevent overfitting. For traditional ML models we used it with grid search and randomised search for automated hyperparameter tuning, and for deep learning models, we implemented it manually with per-fold feature scaling. Stratified K-fold cross-validation was chosen due to the imbalanced nature of our dataset which makes it appropriate for this as per Prusty, Patnaik, and Dash, 2022. It is used to prevent data leakage in imbalanced sets.

StandardScaler was used to normalise numerical features in the dataset except age, to be used by all the models.

- **Linear Support Vector Classifier (SVM):** LinearSVC is an implementation for SVM, well-suited for text classification with high-dimensional sparse features. A grid search strategy was employed over a predefined set of hyperparameters. The optimal configuration was identified through stratified k-fold cross-validation, ensuring consistent class distribution across training and validation folds. This process systematically evaluated combinations of hyperparameters to maximize classification accuracy and generalisability.
- **Random Forest:** For the second model, we employed Random Forest Classifier with randomised search approach to explore efficient hyperparameter space. A total of fifty configurations were evaluated through five-fold cross-validation. Performance was assessed using the standard classification metrics, and class imbalance was addressed by applying balanced class weights. Out-of-bag estimation was incorporated to provide an additional measure of generalization performance.
- **BiLSTM with Attention:** We built a bidirectional LSTM with attention to integrate textual and numeric features. Text inputs were represented using fixed pretrained GloVe embeddings (glove.6B.300D), while the attention layer emphasized informative sequence components. The concatenated representation was passed through a feedforward classifier with dropout. Data was split into training, validation, and test sets using stratified sampling. Model training employed AdamW optimization and cross-entropy loss, with early stopping based on validation performance. Training was performed using Adam and cross-entropy loss, with validation metrics tracked across epochs. Evaluation included confusion matrix analysis and visualization of training dynamics. The weights from the models were saved after evaluation was performed on the test set.
- **BiLSTM without Attention:** For the fourth model, a bidirectional LSTM without Attention was developed to integrate textual and numeric features for binary classification. We use the glove embeddings and the saved weights from the previous model. However, due to a difference in the architecture in the fourth model caused by the absence of Attention layer, we only loaded compatible weights from the previous models which are identified through a compatibility check. The incompatible weights from the previous model are discarded by this model and for the specific layer, a random weight is initialised. Final evaluation included the same metrics on the held-out test set.

A. Feature Importance and Comparative Analysis

We then examined various combinations of text features and numerical features each model deemed important for

TABLE III: Models and Approaches.

Model	Setup
LinearSVC	Tf-Idf+StandardScaler
Random Forest	Tf-Idf+StandardScaler
BiLSTM with Attention	GloVe embeddings+StandardScaler
BiLSTM without Attention	GloVe embeddings+StandardScaler

distinguishing dementia from control. For comparative analysis, we used weighted sum heuristic based on the importance scores of the features. The following formula was used:

$$\text{Combined_Score} = W_t \times \text{importance_value}_t + W_n \times \text{importance_value}_n,$$

where W_t = Weight assigned to feature importance value of text,

and W_n = Weight assigned to feature importance value of numerical feature

For this study, these weights were set manually and were based on practical application and intuition. We assigned a weight of **0.6** to W_t and **0.4** to W_n .

The *Weighted Sum* method is a well-established technique commonly employed in multi-criteria decision-making contexts (Williams and Cai, 2025). This approach is also widely used in search engine ranking algorithms (Luh, Yang, and Huang, 2025). In our comparative analysis, feature coefficients are interpreted as decision criteria, and corresponding weights are derived from prior experimental results. Specifically, we conducted classification tasks using text features and numerical features independently on LinearSVC. The findings indicated that classifiers achieved substantially higher performance when utilizing text features alone compared to numerical features suggesting that text features provide the primary source of information to the models.

1) *LinearSVC (Model Coefficients):* The LinearSVC model provides coefficients that identify the most important features. We applied filters for trigrams and tetragrams ($n=3$ and $n=4$) to capture meaningful phrases. These, along with coefficients for numeric features were extracted. To identify most important feature combinations, we first identified top 10 most important trigrams/tetragrams using SVC's model coefficients. Likewise, model coefficients of numerical features were also extracted. Using combined score, we extracted ten most important feature combinations of text and numerical features. The model coefficients we extracted from LinearSVC point in the directions that distinguish the classes. A coefficient with a negative sign will point toward class 0 and a coefficient with a positive sign will point toward class 1.

2) *Random Forest (Feature Importance):* The Random Forest model provides Gini importance scores that identify features contributing most to classification accuracy through impurity reduction across decision trees. We ap-

plied filters for trigrams and tetragrams (n=3 and n=4) to capture meaningful phrases, extracting their importance scores along with those of numeric features from the trained Random Forest model. To identify most important feature combinations, we identified the top 20 most important trigrams/tetragrams based on their Gini importance rankings. We then computed combined scores of each of these 20 text features and each numeric feature. We extracted the ten highest-scoring combinations.

3) *BiLSTMs (SHAP Values)*: For the BiLSTM model, we first created a model wrapper class to ensure compatibility with KernelExplainer. Using KernelExplainer, we computed SHAP values for a subset of the test data to determine the contribution of each feature to the model's predictions for each class. The SHAP-based model provides explainability scores by attributing additive feature contributions across the neural network's outputs. Feature importance was computed by averaging SHAP values across instances, with n-gram importance calculated as the mean SHAP value across all occurrences of each n-gram in the dataset, and numeric feature importance calculated as the mean SHAP value across all instances for each feature.

To capture meaningful phrases, we applied filters for trigrams and tetragrams (n=3 and n=4), extracting their SHAP values alongside those of numeric features from the trained model wrapper. We then identified the top ten most important feature combinations based on combined score of shap values of the text feature and the numerical feature in the combination. We implement a logic to filter out redundant ngrams by presenting a diverse set of important textual features.

Additionally, we applied a scoring logic where the combined score of each feature pair was multiplied by -1 if it favored classification as 'AD', and by 1 if it favored 'Control'. This approach highlights feature pairs with contrasting SHAP attribution patterns across diagnostic groups. Combinations with positive adjusted scores were categorized as control-discriminative, while those with negative scores were considered AD/MCI-discriminative.

Finally, we extracted the ten highest-scoring feature combinations based on the absolute value of their adjusted scores.

IV. RESULTS

A. Classification Performance

All models achieved robust performance on the dementia classification task. The Random Forest yielded the highest test accuracy (86%), followed by BiLSTM with Attention mechanism (84.75%). The LinearSVC came third by achieving an accuracy of 83% on the test set and the BiLSTM without Attention achieved test accuracy of 77.97%. Precision and recall were balanced for each class, resulting in F1-scores comparable to the accuracies. These results are in line with prior studies: for example, Bouazizi *et al.* reported 83.56% accuracy with an LSTM model on the same corpus.

TABLE IV: Classification performance: Accuracy (Acc), Precision (P), and F1-score by class

Model	Acc (%)	Precision(%)		F1-Score(%)	
		C	D	C	D
LinearSVC	83	79	85	75	87
Random Forest	86	93	84	67	97
BiLSTM+Attn	84.75	80	87	76	89
BiLSTM	79.66	91	77	62	86

C = Control, D = Dementia

B. Feature Combinations

Text Features: The most important phrases captured by LinearSVC are visualized in Figure 3. The model identified distinctive linguistic patterns that differentiate dementia patients from controls.

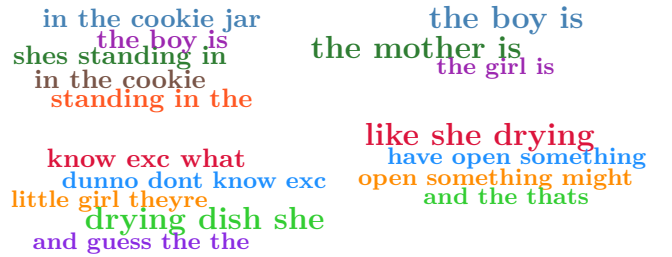


Fig. 3: Comparative visualization of key phrase patterns identified by a) LinearSVC, b) Random Forest, c) BiLSTM with Attn, and d) BiLSTM

TABLE V: Important Feature Combinations

Feature Combination	Score
'in the cookie jar' + 'num_unusual_pos_bigrams'	-0.36
'the boy is' + 'num_fragments'	0.35
'in the cookie jar' + 'num_false_starts'	-0.35
'standing in the' + 'num_fragments'	0.34
'in the cookie' + 'num_unusual_pos_bigrams'	-0.34
'the boy is' + 'num_fillers'	0.34
'standing in the' + 'num_fragments'	0.34
'in the cookie' + 'num_false_starts'	-0.34
'shes standing in' + 'num_fillers'	0.33
'standing in the' + 'num_fillers'	0.33

1) *LinearSVC*: Among the ten most influential feature combinations identified by the LinearSVC model, only five ngrams were present out of which four were trigrams. The combination of the tetragram "in the cookie jar" with the numerical feature num_unusual_pos_bigrams yielded the highest combined score. Notably, four out of the ten combinations exhibited negative combined scores. These combinations were ranked by the magnitude of their scores, regardless of directionality. Another interesting insights was that only four of the eight numerical features namely num_unusual_pos_bigrams, num_fragments, num_false_starts, and num_fillers were present across all ten combinations. Particularly, the pairing of the trigram "in the cookie" or the tetragram

"in the cookie jar" with `num_unusual_pos_bigrams` and `num_false_starts` contributed significantly to the classification of samples within the AD/MCI group. This suggests that samples characterized by frequent use of these specific n-grams, alongside a high occurrence of syntactically atypical bigram constructions, were more likely to be associated with cognitive impairment. Furthermore, `num_unusual_pos_bigrams` and `num_false_starts` were the numerical features consistently aligned with predictions of Alzheimer's disease (AD). This may be indicative that unusual sequences of words and abandoned attempts at phrasing are more likely to suggest cognitive impairment than other linguistic attributes. In contrast, the remaining eight combinations each involving a distinct n-gram paired with `num_fragments` and `num_fillers` were indicative of samples not associated with cognitive impairment. This pattern implies that the LinearSVC model did not consider filler words as a strong indicator of cognitive impairment when measured as a combined feature. Interestingly, what was observed is that `num_fragments` which is a subset of unusual bigrams, specifically focusing on nouns, was associated with the 'Control' group. This may suggest that people with cognitive impairment had high occurrence of unusual word sequences that were more general.

Overall, the model demonstrated a selective reliance on specific linguistic and structural features, with selective ngrams and numerical features.

TABLE VI: Important Feature Combinations

Feature Combinations	Score
'the boy is' + 'num_fillers'	0.0055
'the mother is' + 'num_fillers'	0.0054
'the boy is' + 'num_fragments'	0.0051
'the mother is' + 'num_fragments'	0.0050
'the boy is' + 'num_false_starts'	0.0049
'the mother is' + 'num_false_starts'	0.0048
'the boy is' + 'num_unusual_pos_bigrams'	0.0048
'the mother is' + 'num_unusual_pos_bigrams'	0.0048
'the girl is' + 'num_fillers'	0.0047
'the boy is' + 'num_revisions'	0.0046

2) *Random Forest*: In contrast to the LinearSVC model, which identified a diverse set of n-grams in its most influential feature combinations, the Random Forest model highlighted only three trigrams "the boy is", "the mother is", and "the girl is". All top-ranked combinations involved trigrams exclusively, with no tetragrams present, and each had a positive combined score. The pairing of "the boy is" with the numerical feature `num_fillers` emerged as the most impactful, suggesting that samples with frequent use of this trigram and a high occurrence of filler words were more easily distinguishable. Moreover, Random Forest identified filler words and unusual word sequences both general and noun specific as highly influential in the classification process. However, unlike LinearSVC, the Random Forest model did not provide insight into the directionality of predictions, as its use of

Gini importance scores which being non-negative make it difficult to determine whether a feature combination favors one class over another. Despite having a greater accuracy than the previous model, this model was less interpretable and may require a more comprehensive analysis for identifying class-specific combinations. Furthermore, it may be interpreted that the model was heavily reliant on the trigram structure DT + (human)NN + VBZ for prediction as evident from its presence among the most influential combinations.

TABLE VII: Important Feature Combinations

Feature Combination	Score
'know exc what' + 'num_false_starts'	-0.051
'dunno dont know exc' + 'num_false_starts'	-0.051
'little girl theyre' + 'num_false_starts'	-0.05
'drying dish she' + 'num_fillers'	-0.045
'and guess the the' + 'num_false_starts'	-0.043
'drying dish she' + 'num_repetitions'	-0.041
'drying dish she' + 'num_disagreements'	-0.041
'drying dish she' + 'num_pauses'	-0.041
'know exc what' + 'num_pos_bigrams'	-0.041
'drying dish she' + 'num_revisions'	-0.041

3) *BiLSTM with Attention*: The BiLSTM model with Attention identified five distinct n-grams among its top ten most influential feature combinations. The trigram "drying dish she" appeared most frequently, while the phrase "know exc what" featured in the highest-ranked combination. Unlike the previous models, this one included multiple tetragrams among its most prominent combinations, as illustrated in Figure 3. A notable characteristic of this model is that all top ten feature combinations were associated with the Dementia class. The repeated presence of "drying dish she" suggests that frequent use of this trigram, along with a high occurrence of filler words, repetitions, and pauses, may be indicative of cognitive impairment. Among these, the high frequencies of the trigram and filler words was more predictive than its pairing with repetitions and pauses. However, the exclusive association of all top combinations with the Dementia class may reduce the interpretability of the model. Unlike Random Forest, however, shap model provides directional attribution(although through a simple heuristic), allowing for clearer insights into class-specific feature contributions.

TABLE VIII: Important Feature Combinations

Feature Combination	Score
'like she drying' + 'num_fillers'	-0.041
'like she drying' + 'num_pauses'	-0.038
'like she drying' + 'num_unusual_pos_bigrams'	-0.038
'like she drying' + 'num_fragments'	-0.038
'like she drying' + 'num_repetitions'	-0.037
'like she drying' + 'num_disagreements'	-0.037
'have open something' + 'num_false_starts'	-0.036
'open something might' + 'num_false_starts'	-0.036
'and the thats' + 'num_fillers'	-0.036
'like she drying' + 'num_revisions'	-0.035

4) *BiLSTM without Attention*: This model identified only four distinct n-grams among its most influential fea-

ture combinations, with 'like she drying' emerging as both the most impactful and the most frequently occurring. All eight available numerical features appeared in these combinations. The ngrams identified by this model as influential were all different from the ngrams identified by the attention-based model. However, a noticeable trait was observed in the two BiLSTMs, and that was the presence of the word 'drying' in the most prominent ngrams. This may suggest that the region in the cookie theft picture where a woman is 'drying' the dishes was informative for the BiLSTMs to differentiate people associated with cognitive impairment from those who were healthy. Additionally, all of the eight handcrafted numerical features contributed to some degree in classification task.

An interesting insight from the SHAP analysis was that, for both models, top ten feature combinations were associated with cognitive impairment. This trend aligns with the class imbalance in the dataset, which contains more samples from cognitively impaired individuals than from healthy controls. However, the LinearSVC model exhibited a contrasting behavior: it had more feature combinations in the top ten associated with Control than Dementia, despite the Control samples being fewer than those labeled Dementia. This suggests that LinearSVC may be more sensitive to discriminative features of healthy speech patterns, or potentially overcompensating for the class imbalance in its decision boundary.

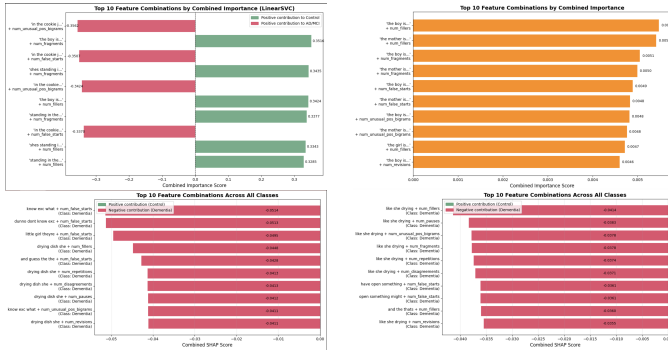


Fig. 4: Comparison of combined importance scores for different feature combinations.

V. DISCUSSION

This study was based on the idea that people who are cognitively impaired exhibit inconsistencies in their speech, which become early markers for detection of dementia. Our approach was to examine these markers through the use of feature combinations to get a better understanding of how models interpret inconsistencies in human speech and if they identify similar combinations and can be interpreted. By analysing the results across the models, we can assess if models process similar information as human clinicians.

Our results showed that LinearSVC, despite being less accurate than Random Forest and BiLSTM with At-

tention, was the most interpretable model. This can be attributed to the nature of its model-specific tool which inherently tells which direction the feature combinations point towards. Conversely, Random Forest, that was the best performing model out of the four, was least interpretable as it only displayed combinations that influenced its predictions, but not the class. BiLSTMs had a great balance of accuracy as well as interpretability, with a simple implementation of a heuristic to make the class distinction visible.

Few limitations of our study, included limiting feature combinations to only trigrams and tetragrams. This was done to ensure a fair comparison between the models. Another limitation of our study, is that we treated the features in the combinations independently. We assessed the frequencies of ngrams and numerical features, and not co-occurrence or interaction. For deep models, we chose the models with seeds that reached accuracy of 80% instead of trying with more seeds.

VI. CONCLUSION

In this study, we performed a comparative analysis of dementia classification models. Unlike other comparative studies which focus on model accuracy and performance, this study focused on comparing the consistency of feature combinations and model interpretability across SVM, Random Forest, and BiLSTMs. We combined text features with handcrafted numerical features and used Model-specific tools for classical models and model-agnostic tools for deep learning models. We used weighted scoring to identify most influential feature combinations and found that all four models identified different feature combinations that they found influential and their interpretability also differed.

VII. FUTURE WORK

The study performed a comparative analysis of classification models based on importance of feature combinations and interpretability. It focused on identifying top ten feature combinations. Future research could extend this work in several promising directions. More combinations could be identified for a more comprehensive analysis of feature importance, model behaviour, and interpretability. While this study concentrated on ten feature combinations, subsequent investigations can study more combinations, might incorporate analyses of feature co-occurrences and interactions to enhance understanding of underlying model dynamics. The current study used weighted scoring heuristic to identify the feature combinations. In the future, different heuristics may be applied and the feature combinations may then be compared for consistency. Furthermore, the evaluation of AI interpretability in this work was achieved through analyses and did not include surveys and feedback from clinical experts, which may be addressed in the future.

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